ScienceDirect

Juncus

In subject area: Agricultural and Biological Sciences

Chapters and Articles

You might find these chapters and articles relevant to this topic.

Review article

Resolving the puzzle of Martin's broad embryo: A solution based on morphology, taxonomy and phylogeny

2018, Perspectives in Plant Ecology, Evolution and Systematics

Carol C. Baskin, Jerry M. Baskin

2.2 Juncaceae

This family also belongs to the Poales, but its embryo differs greatly from that found in the Eriocaulaceae, Mayacaceae and Xyridaceae. Martin (1946) has a drawing of the seeds of *Luzula compestris* and *Juncus marginatus* that he labels as 'broad". However, the embryo in both drawings is taller than wide, which does not fit his definition that a broad embryo is wider than tall. He says "…the Juncaceae seems to serve as an intermediate link [from broad to capitate]." Le Maout and Decaisne (1867 [1873])Le Maout and Decaisne, 1873Le Maout and Decaisne (1867 [1873]) illustrate the embryo of a *Luzula* sp. as being ellipsoidal and that of a *Juncus* sp. as being somewhat capitate (mushroom-like). Laurent (1904) described the embryo of Juncaceae as being like a top on one end and reduced in size and oval on the other end. The cotyledon is physically terminal (Vierhapper, 1930), and the cotyledon and hypocotyl ends of the embryo are separated by a lateral depression (Souèges, 1923, 1933). The seeds of Juncaceae have starchy endosperm (Martin, 1946), and the embryo is surrounded by endosperm (Hamann, 1964). Zaman (1950) described the embryo of *Juncus primatocarpus* and *J. effusus* as

On this page

the cotyledon remains inside the seed, but the emerged-part of the cotyledon develops into an elongated leaf-like structure (phaneromer). The meristem inside the cotyledonary sheath gives rise to new leaves. In addition to the primary root, there are shoot-derived roots (Tillich, 2007).

The embryo in seeds of *Prionium serratum* (previously in the Juncaceae but now in the Thurniaceae or Prioniaceae) is at least three times taller than wide, is embedded in endosperm and

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S1433831918300507

Review article

The potential of saltmarsh halophytes for phytoremediation of metals and persistent organic pollutants: An Australian perspective

2022, Marine Pollution Bulletin

Rebecca A.L. Roe, Geoff R. MacFarlane

5.1.4 Family Juncaceae

Juncus kraussii of the Juncaceae family, is an Australian native rush, also found in New Zealand and South Africa. The rush typically occupies the mid to upper zones of the marsh, often occurring in dense monospecific stands 1–2 m in height forming a landward fringe in the saltmarsh. In the only published paper regarding metal accumulation and partitioning in *J. kraussii*, Stein et al. (2021) reported BCF values >1 for Cd (1.44) and approaching unity for Cu (0.85) but lower root accumulation of Zn (BCF=0.46), Pb (BCF=0.27), Se (BCF=0.24) and As (BCF=0.25). However, once in the root, As, Cu, Pb, and Zn were highly mobile with TF values of 1.33, 1.8, 4.37, and 7.32 respectively (Stein et al., 2021). Therefore, it appears that *J. kraussii* may be utilised for phytostabilisation of Cd and phytoextraction of As, Cu, Pb and Zn from upper saltmarsh sediments, and other contaminated soils, pending further investigations.

Another member of the family Juncaceae, *Juncus bufonius*, has shown promise as a phytostabilising species, accumulating large amounts of Cd (34.34 mg/kg), Cu (251.07 mg/kg), Ni (6.6 mg/kg) and Pb (718.44 mg/kg) to roots when grown on mine tailings for a 4-month period, with mean BCF values of 3.91, 1.79, 18.36, and 0.4 respectively (Chang Kee et al., 2018). Additionally, *J. bufonius* proved to be an efficient phytoextractor species for Zn and more

On this page

species, growing to a height of 20–30cm. *J. bufonius*, is found in all

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S0025326X22004933

Chapter

Diversity and Classification of Flowering Plants

2010, Plant Systematics (Second Edition)

Michael G. Simpson

Juncaceae

Rush family (L. for binder, in reference to use in weaving and basketry). 7 genera/ca. 350 species. (Figure 7.68)

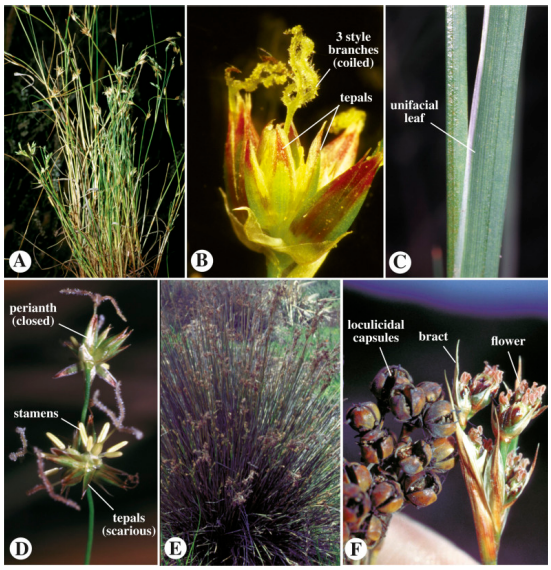

FIGURE 7.68. POALES. Juncaceae. **A**. *Juncus bufonius*, whole plant. **B**. *Juncus dubius*, flower close-up. Note six, imbricate, scarious tepals. **C,D**. *Juncus phaeocephalus*. **C**. Close-up of unifacial leaves. **D**. Flowers, showing exserted styles. **E**. *Juncus acutus*. **E**. Whole plant. **F**. Close-up of infructescence of capsules (left) and inflorescence (right).

Show more

Read full chapter

View PDF

Explore book

Read full chapter

On this page

Halophytic herbs of the Mediterranean basin: An alternative approach to health

2018, Food and Chemical Toxicology

S.A. Petropoulos, ... I.C.F.R. Ferreira

2.9 *Juncus acutus* L.

2.9.1 Botanical aspects

Spiny rush (*Juncus acutus* L. Juncaceae) is a salt tolerant species found at coastal marshes in the Mediterranean basin (Batriu et al., 2015; Fountoulakis et al., 2017). The salt tolerance of the species is associated with proline and Ca^{2+} and Mg^{2+} accumulation which contribute to plant defense mechanisms against salinity stress through osmoregulation (Boscaiu et al., 2011; Hassan et al., 2017b). Hassan et al. (2017a) observed an increase of total phenolic compounds and flavonoids in the aerial parts of the species under high salinity levels (400 mM NaCl), whereas photosynthetic pigments were sensitive to salt-induced oxidative stress and decreased at levels higher than 100 mM NaCl.

2.9.2 Chemical composition

The wetland plants of *Juncus* genus such as *J. acutus*, are considered rich sources of nitrogen-free alkylated phenanthrenoids, while they also contain secondary metabolites of various classes, such as coumarins, sterols, terpenes, stelbenes, carotenoids and so forth (Behery et al., 2013; El-Shamy et al., 2015). In the study of Behery et al. (2013) several compounds with antioxidant activity were isolated from *J. acutus* rhizomes, namely 8,8′-bidehydrojuncusol, juncunol, 5,7-dihydroxychromone and flavone derivatives (apigenin, luteolin, chrysoeriol, luteolin-7-O-β-glucoside and hydnocarpin (Fig. 1c). Plant culms are rich in soluble carbohydrates (myo-inositol, clicerol, sorbitol, fructose, glucose and sucrose) which are used in salt tolerance mechanisms (Gil et al., 2011).

2.9.3 Health effects

Juncus acutus constituents, especially nitrogen-free alkylated phenanthrenoids, exert a pronounced antioxidant activity and biological activities, including anti-inflammatory, cytotoxicity, anti-leukaemic, anti-algal and cognitive-enhancing properties (Behery et al., 2013; Rodrigues et al., 2017). Moreover, Rodrigues

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S0278691518300966

Review article

On this page

halophytes for phytoremediation of metals and persistent organic pollutants: An Australian perspective

2022, Marine Pollution Bulletin

Rebecca A.L. Roe, Geoff R. MacFarlane

5.2.6 Family Juncaceae

A large proportion of research based on the phytoremediation capacity of *Juncus* spp. for organic contaminants concerns the nearly cosmopolitan species *J. effusus* (Syranidou et al., 2017), an introduced species in Australia but often regarded as naturalised in SE Australia and south-western WA. *J. effusus* has shown potential for the phytoremediation of PFAAs (W. Zhang et al., 2019) and pesticides (Bouldin et al., 2006), and as adsorbents of PAHs (Fu et al., 2020) and dyes (Ren et al., 2021). Meanwhile, the Australian native saltmarsh species *J. bufonius* appears to be absent from organic pollutant phytoremediation literature. Another Australian native saltmarsh species, *J. kraussii* has shown some potential for phytoremediation and was utilised in a South African constructed wetlands to aid in removal of OPP pesticides from surface waters in conjunction with other species (*Typha capensis*, and *Cyperus dives*) (Schulz and Peall, 2001). Moreover, *J. kraussii* has proved to be efficient in the removal of total nitrogen, total phosphorous and NaCl from aquaculture effluent (Lymbery et al., 2006). However, the specific phytoremediation capacity of *J.*

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S0025326X22004933

Review article

Halophytic herbs of the Mediterranean basin: An alternative approach to health

2018, Food and Chemical Toxicology

S.A. Petropoulos, ... I.C.F.R. Ferreira

2.9.2 Chemical composition

The wetland plants of *Juncus* genus such as *J. acutus*, are considered rich sources of nitrogen-free alkylated phenanthrenoids, while they also contain secondary metabolites of various classes, such as coumarins, sterols, terpenes, stelbenes, carotenoids and so forth (Behery et al., 2013; El-Shamy et al.,

On this page

8,8′-bidehydrojuncusol, juncunol, 5,7-dihydroxychromone and flavone derivatives (apigenin, luteolin, chrysoeriol, luteolin-7-O-β-glucoside and hydnocarpin (Fig. 1c). Plant culms are rich in soluble carbohydrates (myo-inositol, clicerol, sorbitol, fructose, glucose and sucrose) which are used in salt tolerance mechanisms

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S0278691518300966

Chapter

SUBTRIBE ARTHROSTYLIDIINAE

1999, The Bamboos of the World

D. OHRNBERGER

- Taxonomic and nomenclatural references:

  Glaziophyton Franchet in J. Bot. Paris 3, 1889: 277; type:

  Glaziophyton mirabile Franchet; McClure in Smithson. Contr. Bot. no. 9, 1973: 83

  Arundinaria sect. *Glaziophyton* Hackel in Engler & Prantl, 1897:46
- Tribal assignment: trib. *BAMBUSEAE*, subtrib. *ARTHROSTYLIDIINAE*
- Features: a bamboo of almost total lack of nodes, branches, and leaves; the outer appearance resembles a member of *Juncus.*
- Etymology: The generic name is dedicated to the French civil engineer and botanist Auguste François Marie Glaziou

Read full chapter

View PDF

Explore book

Read full chapter

URL: https://www.sciencedirect.com/science/article/pii/B9780444500205500125

Mini review

Distribution of isoflavonoids in non-leguminous taxa – An update

2006, Phytochemistry

Zuzana Mackova, ... Oldrich Lapcik

On this page

In addition to the six monocot families listed by Reynaud et al. (2005), isoflavones have been reported in two *Juncus* species (Juncaceae) (Abdel-Mogib, 2001) and furthermore in *Aloe vera* (Asphodelaceae) (Saxena and Sangeeta, 2000) and *Smilax glabra* (Smilacaceae) (Yi et al., 1998). Certain confusion exists in the literature concerning taxonomic classification of the last two species. The genus *Aloe* has been classified to the Alooidae subfamily of the Asphodelaceae (Watson and Dallwitz, 1992) or to the Liliaceae (Gutterman and Chauser-Volfson, 2000), alternatively the Aloeaceae has been treated as an independent family (http://plants.usda.gov/, 5th September 2005). The genus *Smilax* has been repeatedly assigned to the Liliaceae as well (Dua et al., 2005; Guo et al., 2004). Most recently, Maver et al. (2005) reported bioassay-guided isolation of six isoflavones with interesting fungicidal activities from the stem of *Eriophorum*

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S0031942206000410

Review article

Resolving the puzzle of Martin's broad embryo: A solution based on morphology, taxonomy and phylogeny

2018, Perspectives in Plant Ecology, Evolution and Systematics

Carol C. Baskin, Jerry M. Baskin

2 Survey of embryo and seed germination morphology

2.1 Eriocaulaceae, Mayacaceae and Xyridaceae

These three monocot families listed by Martin as having a broad embryo belong to the Poales, and their seeds generally have been described by Takhtajan (1997) as having a minute, bell-like, lens-shaped, obovate or globose undifferentiated or weakly differentiated embryo and copious starchy endosperm (Fig. 1A). This embryo description also applies to various other Poales, including the Anarthriaceae (Linder and Rudall, 1993), Centrolepidaceae (Prakash, 1969), Ecdeiocoleaceae (Rudall et al., 2005), Flagellariaceae (Takhtajan, 1997), Joinvilleaceae (Takhtajan, 1997; Kellogg, 2000), Rapateaceae (Venturelli and Bouman,1988) and Restionaceae (Takhtajan, 1997). However, the embryo in seeds of the other Poales, i.e. Bromeliaceae, Juncaceae, Cyperaceae, Thurniaceae, Typhaceae and Poaceae, have organs and are not lens- or bell-shaped (Takhtajan, 1997). Briggs et al. (2014)

On this page

Anarthriaceae and Centrolepidaceae in Restionaceae.

Show more

Read full article

View PDF

Read full article

URL: https://www.sciencedirect.com/science/article/pii/S1433831918300507

Chapter

The Botanical Dance of Death

2011, Advances in Botanical Research

Joanna Kacprzyk, ... Paul F. McCabe

a Hypoxia stress—Aerenchyma formation

Aerenchyma is a tissue composed of a network of interconnected gas conducting intercellular spaces which provide plant roots with oxygen under hypoxic conditions. There are two types of aerenchyma: lysigenous, formed by collapse and programmed death of certain cells in the cortical region of the root to form air-filled cavities (Kawai *et al.*, 1998), and schizogenous, which develops through breakdown of pectic substances in the middle lamellae resulting in cell separation (Laan *et al.*, 1989). In many wetland species, aerenchyma is formed in a constitutive manner (even in dry conditions), being a pre-adaptive mechanism which can be enhanced in case of flooding in species such as rice (Jackson *et al.*, 1985) or *Juncus effuses* (Visser and Bögemann, 2006). Aerenchyma can also be formed in many dryland species where it is induced by adverse environmental conditions like hypoxia, for instance, in maize (Konings, 1982), wheat (Thomson *et al.*, 1990), sunflower (Kawase and Whitmoyer, 1980) and tomato (Kawase, 1981). Formation of lysigenous aerenchyma can be also stimulated by other abiotic stresses which decrease the amount of available oxygen for respiration or the level of available nutrients. This is because aerenchyma development leads to less root respiration/less root organic material input while the same surface area is used for nutrient uptake (Fagerstedt, 2010). Other inducers of lysigenous aerenchyma formation are high temperature, nitrogen (Konings and Verschuren, 1980), phosphorous (Fan *et al.*, 2003) or sulphur deficiencies (Bouranis *et al.*, 2003) or mechanical impedance (He *et al.*, 1996a).

Cell death during lysigenous aerenchyma formation has been investigated in maize (Gunawardena *et al.*, 2001) and hallmark features of PCD were observed during this process including

On this page

Read full chapter

View PDF

Explore book

Read full chapter

URL: https://www.sciencedirect.com/science/article/pii/B9780123858511000044

Related terms:

Halophyte, Typha, Constructed Wetland, Asteraceae, Cyperaceae, Carex, Macrophyte, Salt Marsh, Heavy Metal, Poaceae.

Recommended publications

Ecological Engineering

Journal

Ecosystem Services

Journal

Aquacultural Engineering

Journal

Advances in Ecological Research

Book series

On this page

ELSEVIER

All content on this site: Copyright © 2025 or its licensors and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

RELX™

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page

On this page